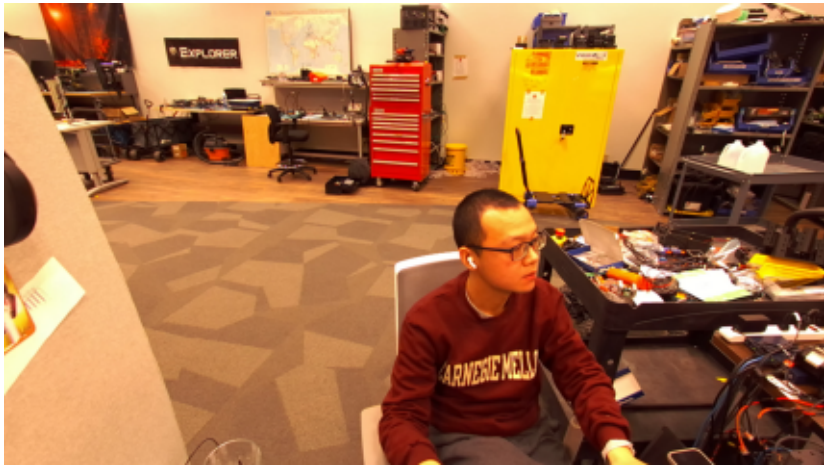
# ZED depth: ONNX vs PyTorch

## PASS: outputs within tolerance

Model: neural_depth_light_5.3.onnx | atol=0.003, rtol=0.003

Left RGB input: im0.png



Right RGB input: im1.png



| Output | Mean abs. | Max abs. | RMSE | P99 abs. | Outside tol. | Result |
|---|---|---|---|---|---|---|
| disparity | 2.92297e-06 | 0.000294209 | 8.18032e-06 | 3.05176e-05 | 0% | PASS |
| disparity_right | 0 | 0 | 0 | 0 | 0% | PASS |
| confidence | 4.34434e-07 | 3.96967e-05 | 1.37292e-06 | 6.4075e-06 | 0% | PASS |

Input HxW: [621, 1104] | network HxW: [256, 512] | bottom/right padding: [0, 0]

Output shape: [1, 1, 128, 256] | comparison includes padding; disparity is not metric depth.

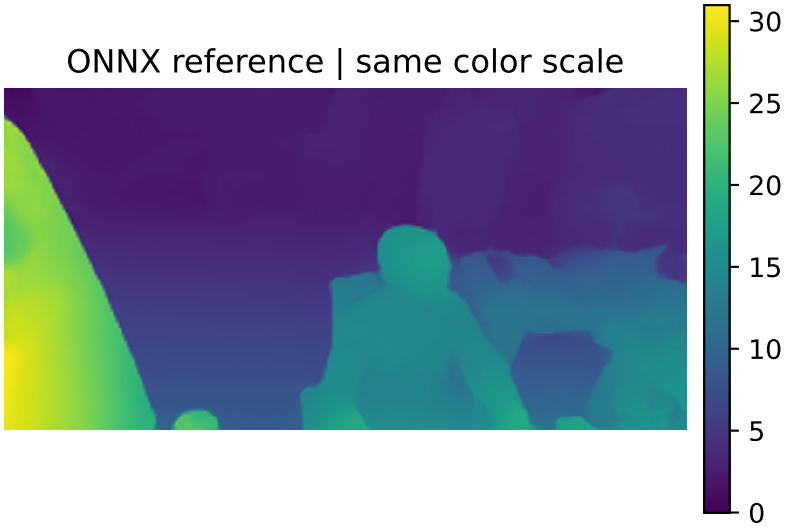
Reference: ONNX Runtime standard layers + independent NumPy custom operators. No PyTorch tensors are injected.

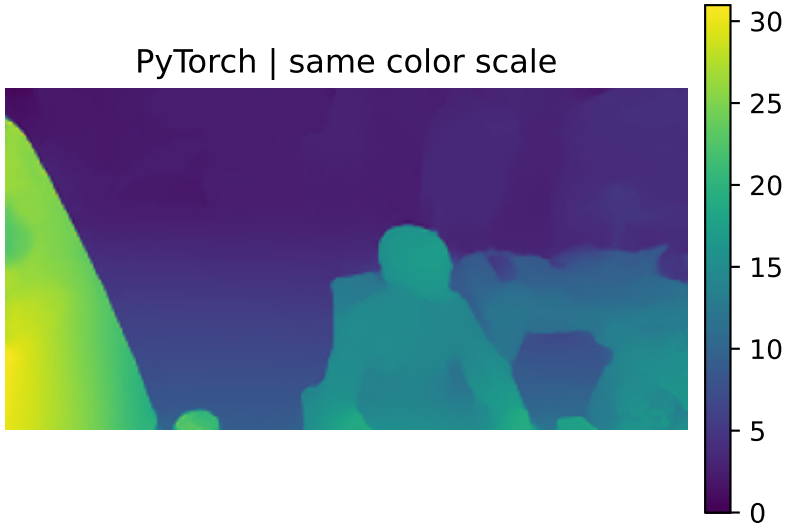
disparity_right is a zero-filled placeholder; matching zeros does not validate right-view depth.

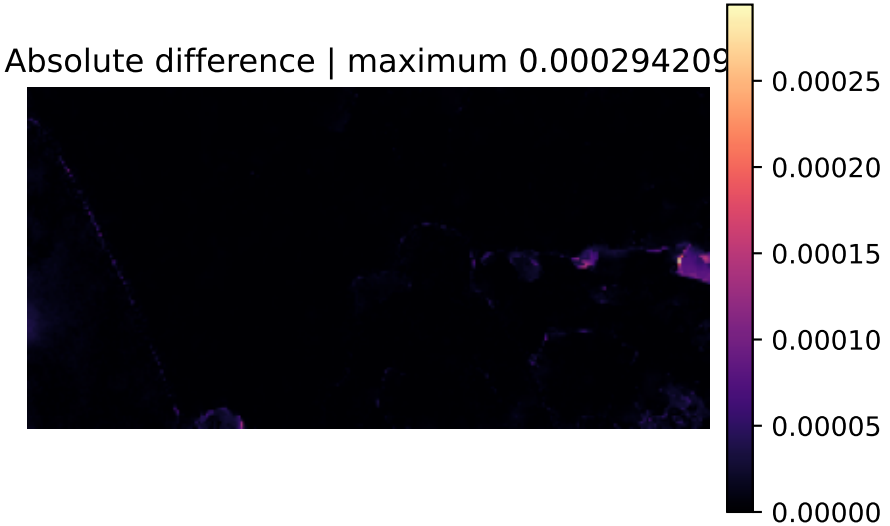
This measures implementation agreement on one stereo pair. No ground truth or native ZED CUDA parity was tested.

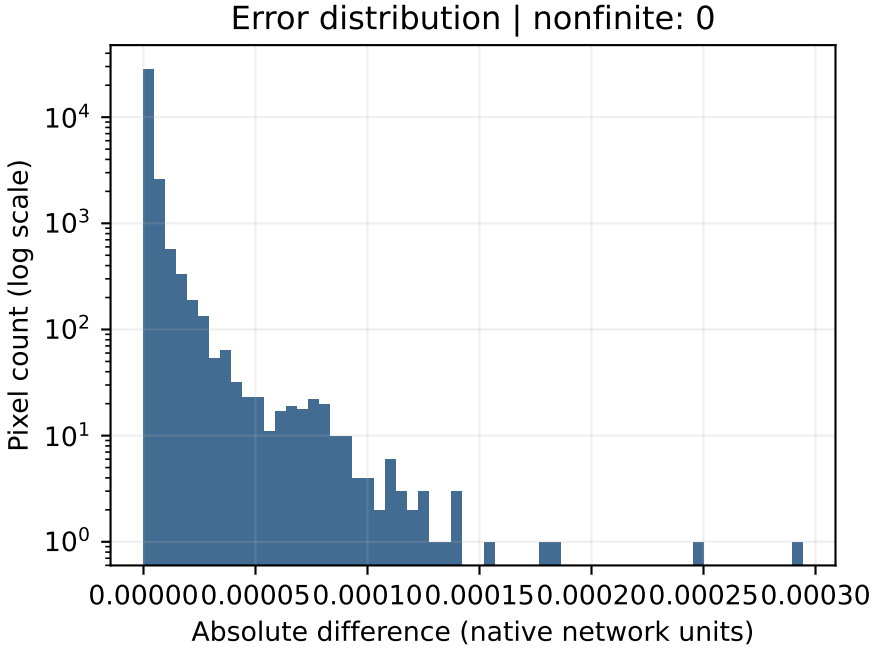
Timing: PyTorch forward 0.128s; reference setup + run 0.120s.

# Disparity output analysis



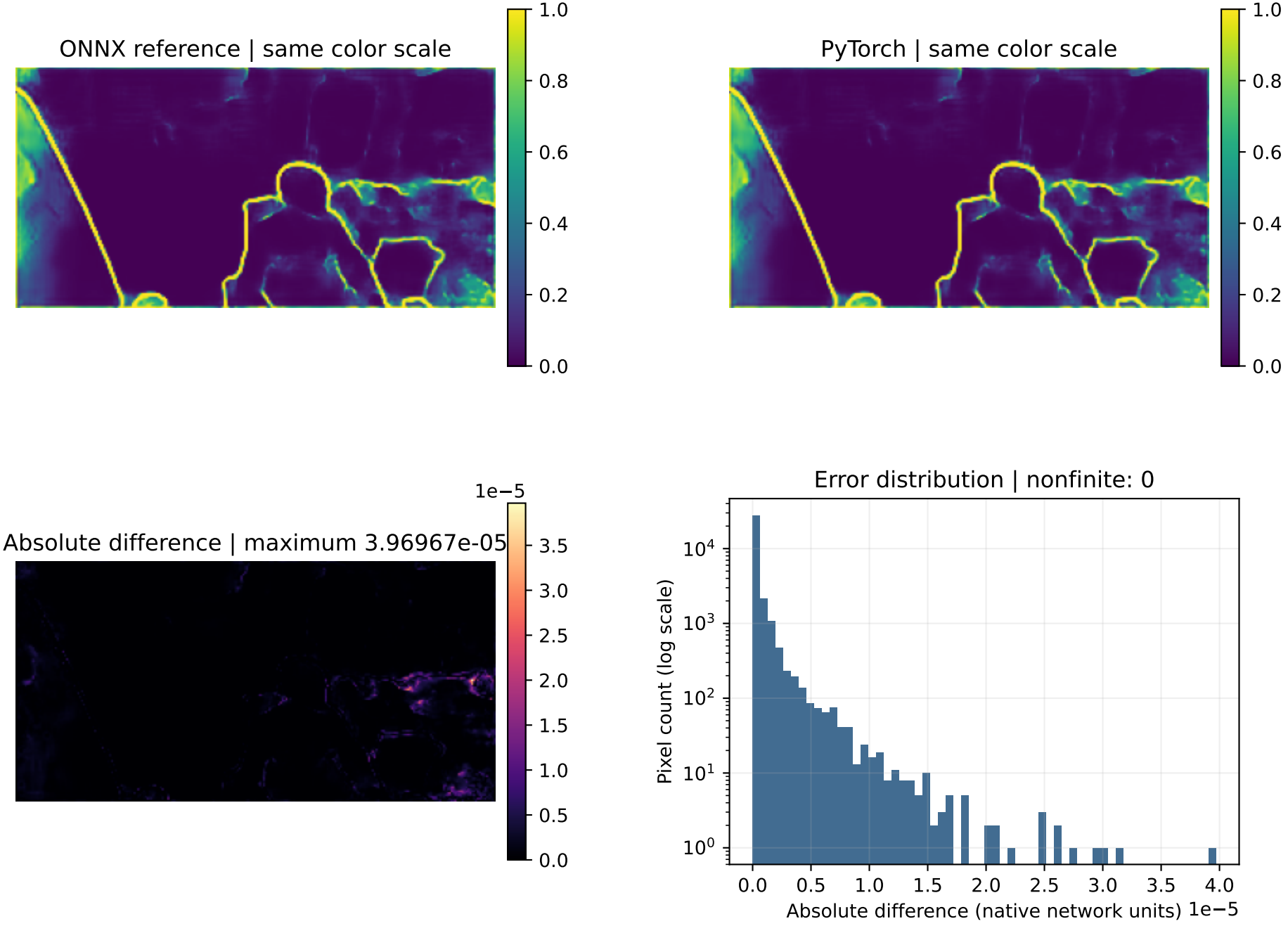




Out-of-tolerance: 0% | Difference colors are magnified to the maximum error; inspect numeric values, not color alone.

# Confidence output analysis




Out-of-tolerance: 0% | Difference colors are magnified to the maximum error; inspect numeric values, not color alone.